

Project Design Phase  
Solution Architecture

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 29 June 2026                                 |
| Name          | HARRI VENDHAN L                              |
| Project Name  | Metro ticket generating system in ServiceNow |
| Maximum Marks | 4 Marks                                      |

Solution Architecture

Solution architecture is a structured process that bridges the gap between passenger requirements and technology solutions. It ensures that the Metro Ticket Booking & Smart Card Recharge System delivers an efficient, secure, and automated digital ticketing experience. Its goals are to:

- **Finds the Best Technology Solution:** Resolves the challenges of manual ticket booking and smart card recharge by implementing a **ServiceNow Service Catalog** with **Flow Designer**, **Business Rules**, and **QR Code generation**, enabling passengers to book tickets quickly and digitally.
- **Describes System Structure & Behavior:** Defines a clear architecture where **Passenger** → **Service Catalog** → **Flow Designer** → **Metro Database** → **QR Code Generation**, ensuring that booking information flows securely and is stored automatically for future verification.
- **Defines Features & Iterative Phases:** Organizes development into logical phases, beginning with metro ticket booking and smart card recharge, followed by automated fare calculation, QR code generation, centralized database management, and future payment gateway integration.
- **Provides Technical Specifications:** Establishes key technical requirements including secure cloud-based data storage, automated workflow processing, QR code generation, role-based access control, reliable record management, and high system availability to support large numbers of metro passengers efficiently.

Example - Solution Architecture Diagram:

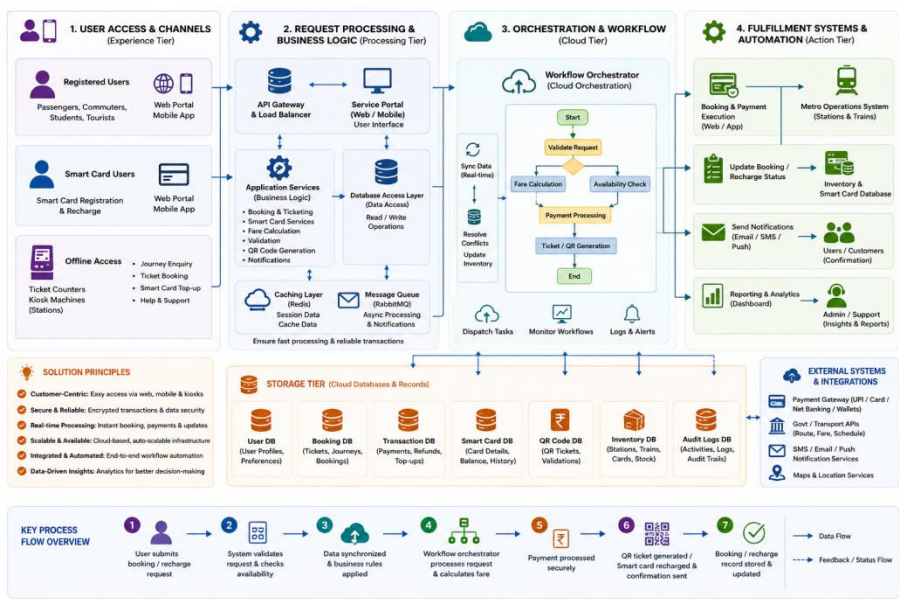


Figure 1: Architecture and data flow of the metro ticket generating sample application

**Reference:**

1. ServiceNow Architecture Center – <https://www.servicenow.com/docs/bundle/latest-architecture-center.html>
2. ServiceNow Flow Designer Documentation – <https://www.servicenow.com/docs/bundle/latest-build-workflows.html>